

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0660

Briscoe Wind 138 kV
June 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0660. The Interconnection Customer proposes a 92.8 MW large load (data center) facility interconnecting at the Briscoe Wind 138 kV substation in the Pennsylvania area. The request was received on August 11, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 335 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$470,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Simpson Stuart Tap-Kempster 138kV line	Medora-Tuco 138kV	115.0	246	5.1
Santee Cooper-Greenland Gap 345kV	Morgantown Energy Facility-Pides Corner 345kV line	105.4	1,700	5.4
Morgantown Energy Facility-Prides Corner 345kV	Greenland Gap-Santee Coper 345kV line	103.3	1,168	10.9
Fishkill Plains Substation-Middle Fork 345kV	Simpson Stuart Tap-Fishkill Plains 345kV Line	103.2	1,547	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Simpson Stuart Tap-Kempster 138kV line	Rebuild line with higher-capacity conductor	\$3,375,000	6% (\$198,000)
Santee Cooper-Greenland Gap 345kV	Reconductor line	\$2,411,000	2% (\$36,000)
Morgantown Energy Facility-Prides Corner 345kV	Reconductor line	\$5,464,000	2% (\$97,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fishkill Plains Substation-Middle Fork 345kV	Reconductor line	\$15,034,000	1% (\$139,000)
	Total allocated to project		\$470,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.